2.49 2nd Order DEs: Hooke's Law

Worked Solutions

1) a)

Extension = current length − natural length, so $x = 1.35 - 1.2 = 0.15$ m.

1) b)

Extension = current length − natural length, so $x = 1.5 - 1.2 = 0.3$ m.

1) c)

Extension = current length − natural length, so $x = 1.1 - 1.2 = -0.1$ m. A negative extension means the string is not stretched. An elastic string cannot push, so it is slack. The extension is taken as 0 m.

2) a)

$T = \frac{\lambda}{l}x$, where $\lambda = 24$ N, $l = 0.8$ m and $x = 1 - 0.8 = 0.2$ m. Hence $T = \frac{24}{0.8} \times 0.2 = 6$ N.

2) b)

$T = \frac{\lambda}{l}x$, where $\lambda = 24$ N, $l = 0.8$ m and $x = 1.3 - 0.8 = 0.5$ m. Hence $T = \frac{24}{0.8} \times 0.5 = 15$ N.

2) c)

Here $x = 0.75 - 0.8 = -0.05$ m, so the string is slack and there is no tension.

3) U)

sing $T = \frac{\lambda}{l}x$ with $T = 8$ N, $l = 2$ m and $x = 2.5 - 2 = 0.5$ m gives $8 = \frac{\lambda}{2} \times 0.5$, so $\lambda = 32$ N.

4) U)

sing $T = \frac{\lambda}{l}x$ with $T = 4.8$ N, $l = 1.5$ m and $\lambda = 18$ N gives $4.8 = \frac{18}{1.5}x$, so $x = 0.4$ m. The length is $1.5 + 0.4 = 1.9$ m.

5) I)

n equilibrium, $R(\uparrow) : T - mg = 0$, so $T = 0.5 \times 9.8 = 4.9$ N. Since $T = \frac{\lambda}{l}x$, $4.9 = \frac{30}{1.2}x$, so $x = \frac{49}{250} = 0.196$ m.

6) a)

In equilibrium, $T = mg = 0.8 \times 9.8 = 7.84$ N. Thus $7.84 = \frac{19.6}{0.6}x$, so $x = \frac{6}{25} = 0.24$ m.

6) b)

Total length $= 0.6 + 0.24 = 0.84$ m.

7) a)

The string has natural length 1 m and current length y, so extension $x = y - 1$. By Hooke's Law, $T = \frac{\lambda}{l}x = \frac{40}{1}(y - 1) = 40(y - 1)$ N.

7) b)

If $y < 1$, the string is shorter than its natural length and is slack. Hence $T = 0$ N.

8) a)

The extension is $x = y - l$.

8) b)

By Hooke's Law, $T = \frac{\lambda}{l}(y - l)$.

8) c)

Taking downwards as positive, weight mg is positive and tension T is negative. Since $a = \frac{dv}{dt} = \frac{d^2y}{dt^2}$, $mg - T = m\frac{d^2y}{dt^2}$, so $m\frac{d^2y}{dt^2} = mg - \lambda\frac{y-l}{l}$.

9) a)

At equilibrium the spring has stretched by e beyond its natural length l, so its length is $l + e$.

9) b)

If the particle moves a further distance x downwards, the spring becomes longer by x. The new length is $l + e + x$.

9) c)

Extension = current length − natural length $= (l + e + x) - l = e + x$.

9) d)

Hooke's Law states $T = \lambda\frac{\text{extension}}{l}$. Since the extension is $e + x$, $T = \lambda\frac{e+x}{l}$.

9) e)

At equilibrium $x = 0$ and acceleration is zero, so the resultant force is zero. The weight mg acts downwards and tension $\lambda\frac{e}{l}$ acts upwards. Thus $mg - \lambda\frac{e+0}{l} = 0$, so $mg = \lambda\frac{e}{l}$.

10) a)

Using Hooke's Law, $T = \lambda\frac{x}{l}$. At equilibrium the extension is e, so $T = \lambda\frac{e}{l}$.

10) b)

In equilibrium the particle has no acceleration, so the resultant force is zero. The weight mg acts downwards while the tension $\lambda\frac{e}{l}$ acts upwards. Hence $mg - \lambda\frac{e}{l} = 0$, so $mg = \lambda\frac{e}{l}$.

10) c)

At equilibrium the extension is e. If the particle moves a further distance x downwards, the total extension is $e + x$.

10) d)

By Hooke's Law, $T = \lambda\frac{\text{extension}}{l} = \lambda\frac{e+x}{l}$.

10) e)

Since x is the downward displacement from equilibrium, $x > 0$ means the particle is below equilibrium. The spring is stretched more than at equilibrium, so tension acts upwards, towards equilibrium. Hence tension makes a negative contribution to the equation of motion.

10) f)

Taking downwards as positive, weight is positive and tension T is negative. Thus $mg - T = m\frac{d^2x}{dt^2}$ and $m\frac{d^2x}{dt^2} = mg - \lambda\frac{e+x}{l}$.

10) g)

From part (f), $m\frac{d^2x}{dt^2} = mg - \lambda\frac{e}{l} - \lambda\frac{x}{l}$. At equilibrium $mg = \lambda\frac{e}{l}$, so $m\frac{d^2x}{dt^2} = -\lambda\frac{x}{l}$.

11) P)

reviously, $m\frac{d^2x}{dt^2} = -\lambda\frac{x}{l}$. Substituting $m = 0.4$, $\lambda = 8$, $l = 0.5$ gives $0.4\frac{d^2x}{dt^2} = -8\frac{x}{0.5} = -16x$, so $\frac{d^2x}{dt^2} = -40x$.

12) a)

Using $m\frac{d^2x}{dt^2} = -\lambda\frac{x}{l}$ with $m = 0.5$, $\lambda = 20$, $l = 0.4$ gives $0.5\frac{d^2x}{dt^2} = -20\frac{x}{0.4} = -50x$, so $\frac{d^2x}{dt^2} = -100x$.

12) b)

The auxiliary equation is $m^2 + 100 = 0$, so $m = \pm 10i$. Thus $x = A\cos 10t + B\sin 10t$. Using $x = 0.1$ at $t = 0$ gives $A = 0.1$. Differentiating, $x' = -\sin 10t + 10B\cos 10t$; using $x' = 0$ at $t = 0$ gives $B = 0$. Hence $x = 0.1\cos 10t$.

13) a)

Using $m\frac{d^2x}{dt^2} = -\lambda\frac{x}{l}$ with $m = 0.3$, $\lambda = 12$, $l = 0.5$ gives $0.3\frac{d^2x}{dt^2} = -12\frac{x}{0.5} = -24x$, so $\frac{d^2x}{dt^2} = -80x$.

13) b)

The auxiliary equation is $m^2 + 80 = 0$, so $m = \pm 4\sqrt{5}i$. Thus $x = A\cos(4\sqrt{5}t) + B\sin(4\sqrt{5}t)$. Using $x = 0.08$ at $t = 0$ gives $A = 0.08$. Differentiating and using $x' = 0$ at $t = 0$ gives $B = 0$. Therefore $x = 0.08\cos(4\sqrt{5}t)$.

14) a)

From the previous result, $m\frac{d^2x}{dt^2} = -\lambda\frac{x}{l}$, so $\frac{d^2x}{dt^2} = -\frac{\lambda}{ml}x$. This is simple harmonic motion with $\omega = \sqrt{\frac{\lambda}{ml}}$.

14) b)

$T = \frac{2\pi}{\omega} = \frac{2\pi}{\sqrt{\lambda/(ml)}} = 2\pi\sqrt{\frac{ml}{\lambda}}$.

15) T)

he force is always directed towards O, so it is opposite in sign to displacement. Since its magnitude is proportional to x, $F = -kx$, where $k > 0$. Using $F = ma$ and $a = \frac{d^2x}{dt^2}$ gives $m\frac{d^2x}{dt^2} = -kx$, so $\frac{d^2x}{dt^2} = -\frac{k}{m}x$. This is simple harmonic motion with $\omega = \sqrt{\frac{k}{m}}$.

16) a)

Take towards the wall as positive. The spring is compressed by x m; its force $T = kx$ opposes the compression and acts negatively. Thus $-kx = m\frac{d^2x}{dt^2}$, so $\frac{d^2x}{dt^2} = -\frac{k}{m}x$. This is simple harmonic motion with $\omega = \sqrt{\frac{k}{m}}$.

16) b)

$T = \frac{2\pi}{\omega} = \frac{2\pi}{\sqrt{k/m}} = 2\pi\sqrt{\frac{m}{k}}$.

16) c)

The particle oscillates about the position where the spring is neither compressed nor stretched, $x = 0$. Here $T = k \times 0 = 0$, so there is no resultant force and this is the equilibrium position.

17) a)

Since $AB = 1$ m, the midpoint O is 0.5 m from each end. At O, each spring has extension $0.5 - 0.4 = 0.1$ m. If displaced a distance x towards B, the left spring length is $0.5 + x$ and its extension $0.1 + x$; the right spring length is $0.5 - x$ and its extension $0.1 - x$. Their tensions are $c(0.1 + x)$ and $c(0.1 - x)$. The left spring pulls towards A (negative) and the right towards B (positive), so $F = c(0.1 - x) - c(0.1 + x) = -2cx$ N.

17) b)

Using $F = ma$, $-2cx = M\frac{d^2x}{dt^2}$, so $\frac{d^2x}{dt^2} = -\frac{2c}{M}x$. This is SHM with $\omega = \sqrt{\frac{2c}{M}}$.

17) c)

$T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{M}{2c}}$.

18) a)

Since $PQ = 1$ m, the midpoint M is 0.5 m from each end. Each spring has extension $0.5 - 0.3 = 0.2$ m at the midpoint. When the particle is at O, the length of each spring is 0.5 m. If the particle is displaced x from M, the spring extensions are $0.2 + x$ and $0.2 - x$, giving tensions $160(0.2 + x)$ and $160(0.2 - x)$. The left spring pulls towards P and the right towards Q, so $F = 160(0.2 - x) - 160(0.2 + x) = -320x$ N.

18) b)

Using $F = ma$, $-320x = 8\frac{d^2x}{dt^2}$, so $\frac{d^2x}{dt^2} = -40x$. This is SHM with $\omega = \sqrt{40} = 2\sqrt{10}$.

18) c)

$T = \frac{2\pi}{2\sqrt{10}} = \frac{\pi}{\sqrt{10}} = \frac{\pi\sqrt{10}}{10}$.

18) d)

The equation $x'' = -40x$ gives $x = A\cos(2\sqrt{10}t) + B\sin(2\sqrt{10}t)$. Using $x = 0.1$ at $t = 0$ gives $A = 0.1$. Differentiating and using $x' = 0$ at $t = 0$ gives $B = 0$. Therefore $x = 0.1\cos(2\sqrt{10}t)$.

18) e)

When $x = 0.04$, $0.04 = 0.1\cos(2\sqrt{10}t)$, so $t = \frac{\arccos 0.4}{2\sqrt{10}} = 0.183298180\ldots$. Since $v = x' = -0.1(2\sqrt{10})\sin(2\sqrt{10}t)$, the speed is $\sqrt{\frac{42}{125}} = 0.580$ m s^{-1} (3 SF). Alternatively, using $v^2 = \omega^2(a^2 - x^2)$ gives $v^2 = 40(0.1^2 - 0.04^2) = \frac{42}{125}$ and $|v| = 0.580$ m s^{-1}.

19) a)

At equilibrium, extension $= 0.5 - 0.3 = 0.2$ m. The tension balances the weight, so $T = mg = 9.8$ N. Since $T = ke$, $9.8 = k(0.2)$ gives $k = 49$. Hence $T = 49e$.

19) b)

At time t, the bob is x m below P, so the spring length is x and extension $x - 0.3$. Thus $T = 49(x - 0.3)$. Taking downwards positive, weight is positive, tension negative, and resistance opposes motion. Since $v = \frac{dx}{dt}$,

$$9.8 - 49(x - 0.3) - 14\frac{dx}{dt} = \frac{d^2x}{dt^2} \quad \Longrightarrow \quad \frac{d^2x}{dt^2} + 14\frac{dx}{dt} + 49x = 24.5.$$

19) c)

The homogeneous equation has auxiliary equation $m^2 + 14m + 49 = 0$, so $(m + 7)^2 = 0$. Its complementary function is $x = Ate^{-7t} + Be^{-7t}$. A constant particular integral $x = c$ gives $49c = 24.5$, so $c = 0.5$. Using $x = 0.8$ at $t = 0$ gives $B = 0.3$. Differentiating, $x' = -7Ate^{-7t} + Ae^{-7t} - 2.1e^{-7t}$. Using $x' = 0$ at $t = 0$ gives $A = 2.1$. Thus $x = 2.1te^{-7t} + 0.3e^{-7t} + 0.5$.

19) d)

$x|_{t=0.8} = 1.98e^{-5.6} + 0.5 = 0.50732\ldots = 0.51$ (2 SF). The equilibrium position is $x = 0.5$ m, so the bob has almost reached equilibrium.

19) e)

As $t \to \infty$, $e^{-7t} \to 0$ and $te^{-7t} \to 0$, so $x \to 0.5$. The bob approaches $x = 0.5$ and does not oscillate.

20) a)

At time t, the particle has moved x m downwards from its initial position. The spring extension is $0.25 + x$, so $T = 16m(0.25 + x)$. Taking downwards positive, $mg = 9.8m$, tension acts upwards and resistance opposes motion. Thus

$$9.8m - 16m(0.25 + x) - 8mv = m\frac{d^2x}{dt^2} \quad \Longrightarrow \quad x'' + 8x' + 16x = 5.8.$$

The auxiliary equation for the homogeneous equation is $m^2 + 8m + 16 = 0$, so $(m + 4)^2 = 0$ and the complementary function is $x = Ate^{-4t} + Be^{-4t}$. A constant particular integral $x = c$ gives $16c = 5.8$, so $c = \frac{29}{80}$. Using $x = 0$ at $t = 0$ gives $B = -\frac{29}{80}$. Differentiating and using $x' = 0$ at $t = 0$ gives $A = -\frac{29}{20}$. Hence $x = -\frac{29}{20}te^{-4t} - \frac{29}{80}e^{-4t} + \frac{29}{80}$.

20) b)

The auxiliary equation has one repeated real root, so the motion is critically damped.

21) a)

The standard form is $T = ke$, where k is stiffness. Therefore stiffness is 6 N m^{-1}.

21) b)

The extension is $2.3 - 2 = 0.3$ m, so $T = 6(0.3) = 1.8$ N.

21) c)

Given $T = 9$ N, $9 = 6e$ gives $e = \frac{3}{2}$ m. The string length is $2 + \frac{3}{2} = 3.5$ m.

22) a)

Let the equilibrium position be E, and take displacement towards R as positive. Since $ER + LE = 10$, $ER = 10 - LE$. The cord extensions are $LE - 3$ and $8 - LE$. At equilibrium the tensions are equal: $2d(LE - 3) = 6d(8 - LE)$, giving $LE = \frac{27}{4} = 6.75$ m. If displaced x m right of equilibrium, the extensions are $\frac{15}{4} + x$ and $\frac{5}{4} - x$, with tensions $2d(\frac{15}{4} + x)$ and $6d(\frac{5}{4} - x)$. The left cord pulls left and the right cord pulls right, so

$$6d\left(\frac{5}{4} - x\right) - 2d\left(\frac{15}{4} + x\right) = 4d\frac{d^2x}{dt^2} \quad \Longrightarrow \quad x'' = -2x.$$

This is SHM with $\omega = \sqrt{2}$.

22) b)

The angular frequency is $\omega = \sqrt{2}$.

22) c)

The equation $x'' = -2x$ gives $x = A\cos(\sqrt{2}t) + B\sin(\sqrt{2}t)$. Using $x = 0.2$ at $t = 0$ gives $A = 0.2$; using $x' = 0$ at $t = 0$ gives $B = 0$. When $x = 0.12$, $t = \frac{\arccos(3/5)}{\sqrt{2}} = 0.6556967\ldots$. The speed is $|x'| = \sqrt{\frac{32}{625}} = \frac{4\sqrt{2}}{25} = 0.226$ m s^{-1} (3 SF). Alternatively $v^2 = 2(0.2^2 - 0.12^2) = \frac{32}{625}$.

23) a)

If $LQ = 2.4$ m, then $QR = 5 - 2.4 = 2.6$ m. The extensions are $2.4 - 1.2 = 1.2$ m and $2.6 - 1.8 = 0.8$ m. The tensions are $18m(1.2) = 21.6m$ N and $27m(0.8) = 21.6m$ N, so they are equal.

23) b)

At time t, $LQ = 2.4 + x$ and $QR = 2.6 - x$. The extensions are $1.2 + x$ and $0.8 - x$, so the tensions are $21.6m + 18mx$ and $21.6m - 27mx$. The normal reaction is $R = mg$, so friction is $0.5mg = 4.9m$. Take positive direction towards R. The block moves left, so friction acts right. Hence

$$(21.6m - 27mx) + 4.9m - (21.6m + 18mx) = mx'' \quad \Longrightarrow \quad x'' + 45x = 4.9.$$

23) c)

In the smooth channel there is no friction, but resistive force $8mv = 8mx'$ opposes motion. Thus $(21.6m - 27mx) - (21.6m + 18mx) - 8mx' = mx''$, giving $x'' + 8x' + 45x = 0$. The auxiliary equation $m^2 + 8m + 45 = 0$ gives roots $-4 \pm \sqrt{29}i$. Hence $x = e^{-4t}(A\cos(\sqrt{29}t) + B\sin(\sqrt{29}t))$. Since $LQ = 2.8$ at release and $LQ = 2.4 + x$, $x = 0.4$ at $t = 0$, so $A = 0.4$. Differentiating and using $x' = 0$ at $t = 0$ gives $B = \frac{1.6}{\sqrt{29}} = \frac{8\sqrt{29}}{145}$. Therefore

$$x = e^{-4t}\left(0.4\cos(\sqrt{29}t) + \frac{8\sqrt{29}}{145}\sin(\sqrt{29}t)\right).$$

24) a)

After displacement x down the slope, the upper-string extension is $a+x$ and the lower-string extension is $b - x$.

24) b)

The component of the weight parallel to the plane is $mg\sin\theta$.

24) c)

The upper-string tension $k_1(a+x)$ acts up the slope, the lower-string tension $k_2(b-x)$ acts down the slope, and the weight component $mg\sin\theta$ acts down the slope. Thus $k_2(b-x) + mg\sin\theta - k_1(a+x) = m\frac{d^2x}{dt^2}$.

25) a)

Let equilibrium be E and let $ME = p$, so $EN = 4.2 - p$. When displaced x m down the wire, the upper and lower string lengths are $p + x$ and $4.2 - p - x$, giving extensions $p + x - 1.5$ and $3.3 - p - x$. Their tensions are $8(p + x - 1.5)$ and $12(3.3 - p - x)$. Take positive direction down the slope; lower tension and the weight component are positive, upper tension negative. Thus

$$\begin{aligned} 12(3.3 - p - x) + 0.5(9.8)\sin 30^\circ - 8(p + x - 1.5) &= 0.5\ddot{x}, \\ 39.6 - 12p - 12x + 2.45 - 8p - 8x + 12 &= 0.5\ddot{x}. \end{aligned}$$

At equilibrium $x = 0$ and $\ddot{x} = 0$, so $20p = 54.05$ and $p = \frac{1081}{400}$. Substitution gives $-20x = 0.5\ddot{x}$, hence $\ddot{x} = -40x$. This is SHM with $\omega = 2\sqrt{10}$.

25) b)

With resistance, $-20x - c\dot{x} = 0.5\ddot{x}$, so $\ddot{x} + 2c\dot{x} + 40x = 0$. The auxiliary equation is $m^2 + 2cm + 40 = 0$. Critical damping requires $c^2 - 40 = 0$, so $c = 2\sqrt{10}$ since $c > 0$.

25) c)

The repeated root is $m = -2\sqrt{10}$, so $x = Ate^{-2\sqrt{10}t} + Be^{-2\sqrt{10}t}$. Since the bead starts 0.15 m below equilibrium, $B = 0.15$. Using $\dot{x} = 0$ at $t = 0$ gives $A = \frac{3\sqrt{10}}{10}$. Thus $x = \frac{3\sqrt{10}}{10}te^{-2\sqrt{10}t} + 0.15e^{-2\sqrt{10}t}$.

25) d)

Differentiating, $v = -6te^{-2\sqrt{10}t}$ and $a = 12\sqrt{10}te^{-2\sqrt{10}t} - 6e^{-2\sqrt{10}t}$. Maximum speed occurs when $a = 0$, giving $t = \frac{1}{2\sqrt{10}}$. Then $v = -\frac{3\sqrt{10}}{10e}$, so greatest speed is $\frac{3\sqrt{10}}{10e}$ m s^{-1}.

26) a)

Let e be the equilibrium extension. The weight component down the slope is $1 \times 10 \sin 45° = 5\sqrt{2}$ N. The tension is $T = \frac{20}{2}e = 10e$. At equilibrium $10e = 5\sqrt{2}$, so $e = \frac{\sqrt{2}}{2}$ m.

26) b)

At time t, the extension is $e + x$ and the tension is $10(e + x)$. Taking down the slope positive, weight acts down and tension and resistance act up. Thus $5\sqrt{2} - 10(e + x) - 2x' = x''$. Using $e = \frac{\sqrt{2}}{2}$ gives $x'' + 2x' + 10x = 0$.

26) c)

The auxiliary equation is $m^2 + 2m + 10 = 0$, so $m = -1 \pm 3i$. Thus $x = e^{-t}(A \cos 3t + B \sin 3t)$. Using $x = 0.2$ at $t = 0$ gives $A = \frac{1}{5}$. Differentiating and using $x' = 0$ at $t = 0$ gives $B = \frac{1}{15}$. Hence $x = e^{-t}(\frac{1}{5} \cos 3t + \frac{1}{15} \sin 3t)$.

26) d)

Differentiating gives $x' = -\frac{2}{3}e^{-t} \sin 3t$. The first time after release with $x' = 0$ is $3t = \pi$, so $t = \frac{\pi}{3}$. Then $x = -\frac{1}{5}e^{-\pi/3}$. At the first turning point, the particle is $\frac{1}{5}e^{-\pi/3}$ m up the slope from equilibrium.